Shizhen Jia

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Objective

Ph.D. candidate in wireless communications at NYU WIRELESS seeking internship opportunities in wireless systems and signal processing. Experienced in MIMO beamforming, interference nulling, ray tracing, game-theoretic optimization, FR3 multi-band channel estimation, and OTA phase calibration, with interests in link-level modeling, FR3 spectrum studies, and scalable terrestrial/non-terrestrial communication systems.

Education

- **New York University** *Sep. 2024 – Present*
  Ph.D. candidate in Electrical Engineering, Advisor: Sundeep Rangan New York
- **New York University** *Sep. 2022 – Aug. 2024*
  M.S. in Electrical Engineering New York
- **Chongqing University** *Sep. 2018 – Jun. 2022*
  B.Eng. in Electrical Engineering China

Experience

- **CableLabs — NTN Study Intern** *May – Aug. 2026*
  Louisville, Colorado, United States | Tools: MATLAB, Python
  - Reproduced 3GPP TR 38.863 NTN–TN coexistence results and extended the simulations to multi-satellite systems; developed tools for LEO spatial reuse and multi-satellite NTN–NTN interference evaluation.
  - Developed satellite scheduling algorithms and evaluated capacity, interference, and throughput-loss trade-offs to support spectrum-sharing studies.
  - Built Starlink telemetry and measurement dashboards using gRPC, latency/throughput tests, route diagnostics, and TLE-based serving-satellite estimates; tested RadioHound hardware and investigated Wi-Fi antenna orientation and unlicensed-band NTN operation.

Research Projects

- **Intra-Cell Spatial Reuse in LEO Downlinks: Beam-Leakage-Aware Satellite Scheduling** *2026*
  NYU WIRELESS / CableLabs | Tools: MATLAB, Python, satellite ephemerides
  - Modeled concurrent Ku-band downlinks with realistic satellite geometry, receive-array leakage, antenna patterns, and atmospheric losses; evaluated a 24-hour Starlink trace with 720 snapshots.
  - Developed beam-leakage-aware branch-and-bound scheduling with an admissible upper bound and greedy-plus-swap initialization, certifying optimality for the reference-point model.
  - Improved simulated mean per-beam capacity by 27.2% over nearest-satellite selection with eight beams; reduced mean search time from 529.48 to 1.98 ms when selecting eight of 28 candidates.

- **NTN–NTN Downlink Adjacent-Channel Coexistence** *May – Aug. 2026*

  CableLabs | Tools: MATLAB, Python | Reference: 3GPP Feature Lead and Way Forward since Dec. 2025

  - Built a 2 GHz multi-satellite LEO–LEO / LEO–GEO downlink interference simulator with seven-beam satellite patterns, configurable geometry, 3GPP propagation, and ITU-R P.618 atmospheric loss.
  - Swept satellite geometry and adjacent-channel interference ratio (ACIR) to quantify aggregate interference from multiple satellites and the resulting SINR and throughput degradation, using a 5% loss criterion.
- **NTN–TN Coexistence Across Six Interference Scenarios** *May – Aug. 2026*

  CableLabs | Tools: MATLAB, Python | Reference: 3GPP TR 38.863

  - Reproduced 3GPP TR 38.863 coexistence results and extended the framework to multi-satellite systems with aggregate interference.
  - Implemented six satellite–terrestrial DL, UL, and cross-link scenarios with urban/rural deployments, 19-site/57-sector wrap-around networks, and multi-beam LEO/GEO configurations.
  - Integrated transmitter leakage, receiver selectivity, uplink resource allocation, and power control; used Monte Carlo and ACIR sweeps to evaluate mean and fifth-percentile throughput loss against a 5% criterion.
- **Non-Coherent Terrestrial–Satellite Downlink Co-Existence and Large-Scale Simulation** *Jan. 2026 – Present*

  Advisor: Sundeep Rangan | Tools: Python, Sionna-RT

  - Study non-coherent terrestrial–satellite downlink co-existence with MUSIC channel estimation.
  - Derive a GLRT for single-user non-coherent angle detection.
  - Analyze INR upper bounds under estimation error and quantify nulling sensitivity.
  - Build large-scale site-specific simulations (20×20 km rural area with 16 base stations).
- **Terrestrial–Satellite Downlink Co-Existence via Joint Detection, Channel Estimation, and Interference Nulling** *Sep. 2024 – Jun. 2025*

  Advisor: Sundeep Rangan | Tools: Python, Sionna-RT

  - Design a 4-stage protocol: uplink broadcast, victim detection, channel estimation, downlink nulling.
  - Develop regularized null-steering beamforming via principal-eigenvector solution (gain–null trade-off).
  - Evaluate overhead and INR/SNR trade-offs using site-specific ray tracing in rural macrocell layouts.
  - Demonstrate large INR reduction vs. no-nulling baseline and improved protection with larger MIMO array.
- **Distributed Uplink Anti-Jamming in LEO Mega-Constellations** *May 2025 – Nov. 2025*

  Advisor: Sundeep Rangan | Tools: Python, Sionna-RT

  - Formulate uplink anti-jamming as a min–max game over transmit covariance matrices (TX vs. jammer).
  - Build a fast solver: water-filling best response (desired TX) + projected-gradient updates (jammer) to the Nash equilibrium.
  - Implement Starlink-geometry case studies with ray-tracing and realistic interference/jammer models.
  - Show 3–5-satellite cooperation improves capacity vs. single-satellite links under close jamming.
- **OTA Phase Measurements: Multipath Impact & Narrowband Array Calibration** *Aug. 2025 – Present*

  Advisor: Sundeep Rangan | Collaborator: Andrea Bedin | Tools: Pi-Radio Marie Curie, RFSoC, MATLAB, Python

  - Study multipath-induced phase errors under MHz-level bandwidths and characterize when errors remain small.
  - Develop a one-shot statistical model linking multipath amplitude to phase-error distribution.
  - Prove multipath phase error is a zero-mean distribution, enabling spatial averaging via controlled antenna movement.
  - Quantify spatial correlation via inter-/intra-measurement statistics vs. displacement and convergence.
  - Propose a low-complexity OTA calibration method using movement + averaging for narrowband array.

- **DMC Multi-band Channel Estimation in FR3 Band** *Dec. 2025 – Present*

  Advisor: Sundeep Rangan | *Tools: Pi-Radio Marie Curie, RFSoC, MATLAB, Python*

  ◦ Implement Newtonized OMP (NOMP) for FR3 multi-band DMC channel estimation (delay/angle refinement).

  ◦ Prepare OTA multi-band experiments on the Pi-Radio/RFSoC platform to validate DMC models.
- **Interference Avoidance for FMCW Automotive Radars via Non-Cooperative Game** *Aug. 2023 – Aug. 2024*

  M.S. Thesis | *Advisor: Quanyan Zhu* | *Tools: MATLAB, Python*

  ◦ Developed a system-level interference framework for FMCW automotive radar networks.

  ◦ Built an analytical collision model linking waveform timing to SINR degradation.

  ◦ Formulated spectrum coordination as a non-cooperative game and derived mixed-strategy equilibria.

  ◦ Quantified robustness gains of game-theoretic policies over heuristic baselines via simulation.

Publications

[C.1] **Shizhen Jia**, Mingjun Ying, Marco Mezzavilla, Doru Calin, Theodore S. Rappaport, Sundeep Rangan (2025). **Joint Detection, Channel Estimation and Interference Nulling for Terrestrial–Satellite Downlink Co-Existence in the Upper Mid-Band**. *Proc. IEEE GLOBECOM*, accepted, Dec. 2025.

[C.2] **Shizhen Jia**, Mingjun Ying, Marco Mezzavilla, Theodore S. Rappaport, Sundeep Rangan (2026). **Distributed Uplink Anti-Jamming in LEO Mega-Constellations via Game-Theoretic Beamforming**. *Proc. IEEE ICC*, accepted, 2026.

[C.3] **Shizhen Jia**, C. Nicolas Barati, Marco Mezzavilla, Sundeep Rangan (2026). **Noncoherent Detection and Interference Nulling for Terrestrial–Satellite Downlink Coexistence in the Upper Mid-Band**. *Proc. IEEE SPAWC*, 2026.

Professional Activities

- **Session Chair** *Dec. 2025*

  IEEE GLOBECOM 2025

  ◦ Channel Modeling & Estimation (SAC-SSC-03), December 10, 2025.
- **Journal Reviewer**

  ◦ IEEE Communications Letters

  ◦ IEEE Transactions on Wireless Communications (TWC)

  ◦ IEEE Transactions on Cognitive Communications and Networking (TCCN)
- **Conference Reviewer**

  ◦ IEEE Global Communications Conference (GLOBECOM)

  ◦ IEEE International Conference on Communications (ICC)